

Less hungry magnets for the experiments of the future

A prototype superconducting coil developed for the SHiP experiment opens the way for more energy-efficient electromagnets for certain applications



View of the demonstrator magnet for the SHiP experiment. (Image: Axelle Collioud, Emma Tommasi/CERN)

How can we advance cutting-edge research but consume less energy? CERN's scientists are working on innovative solutions, and [superconductivity](#) is one of the key ingredients.

A team has recently successfully tested a demonstrator magnet coil that will significantly reduce the power consumption of certain experiments. The coil is made of magnesium diboride (MgB_2) superconducting cables, which are used in the [high-intensity electrical transfer line](#) that will power the High-Luminosity LHC (HL-LHC), the successor to the LHC. It is mounted in a low-carbon steel magnetic yoke that holds and concentrates the field lines, in a so-called superferric configuration.

This innovative magnet is intended for the [SHiP](#) experiment, which is designed to detect very weakly interacting particles and is scheduled to be commissioned in 2031. One of the detector's two magnets must produce a field of approximately 0.5 tesla. The field is of moderate intensity but must be produced in a huge volume that is 6 metres high and 4 metres wide and deep. A normal-conducting resistive electromagnet would have an electrical power of over one megawatt and, as it would have to operate continuously, its power consumption would be high.

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A Word from Luciano Musa

Coming together to celebrate CERN70

This year has seen a wealth of activities for CERN's 70th anniversary. More is to still to come, including the CERN70 Community Event on 17 September

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A Word from Luciano Musa

Coming together to celebrate CERN70

This year has seen a wealth of activities for CERN's 70th anniversary. More is to still to come, including the CERN70 Community Event on 17 September

CERN's 70th anniversary is a remarkable milestone, and celebrations at CERN and across Member and Associate Member States have been taking place since the [official launch](#) in January 2024.

Naturally, more activities are in store. [Registrations have now opened for the dedicated CERN70 Community Event](#) on Tuesday, 17 September on the CERN Meyrin site. The event includes music, food and drink, and of course various forms of entertainment, and all CERN badge holders, including family members and retirees, can register to attend. It will be a perfect occasion to come together to celebrate the achievements of CERN and its international community.

CERN will turn 70 on 29 September, and on 1 October an official ceremony with invited heads of state will honour CERN's legacy as well as look towards to an inspiring future of science and innovation. The ceremony will be broadcast in various locations around the CERN site to enable the community to watch it live.

For those leaving the Geneva area this summer, check out the CERN70 [exhibition at Geneva Airport](#) and the [many events taking place throughout the world](#).

Activities that have already taken place across various regions have had a remarkable impact by

engaging with diverse communities and fostering scientific curiosity in younger generations. These events not only shine a light on CERN's achievements, but also underscore CERN's extraordinary spirit of collaboration across borders, its sharing of knowledge and its unwavering commitment to scientific excellence, training and education.

[Videos and recordings of the CERN Science Gateway public events](#) are now available. This [series of events](#) has highlighted a range of topics across the four main themes of CERN70: pushing back the frontiers of knowledge; advancing science and technology; science without borders; and science for all.

As coordinator of the 70th anniversary celebrations, I find it immensely rewarding to be leading this initiative. I would like to thank everyone involved in the activities so far and the activities still to come. Together we not only look back at groundbreaking discoveries and technological innovations, but also look forward to the ambitions and bright future of particle physics, with many fundamental mysteries yet to be unveiled and new horizons to explore.

*Luciano Musa
Coordinator of the CERN 70th
anniversary activities*

>>> *Continued from page 1*

Hence the idea of using a superconductor that conducts electricity without resistance and thus without energy loss from heating. This is the principle behind the LHC magnets. However, they are based on a niobium–titanium alloy, which requires them to be cooled to a very low temperature of -271 °C (2 kelvin) using superfluid helium produced by a complex cryogenic plant.

Magnesium diboride cables have the advantage of being superconducting at -253 °C (20 kelvin). They can be cooled using gaseous helium and therefore require a less complex cryogenic system, thus offering better thermodynamic efficiency. They could not be used for accelerator magnets such as those of the LHC, which generate fields of around 8 tesla. However, they are suitable for a large magnet with a moderate field like that of SHiP. Built last September, the 1-metre-long demonstrator coil has just successfully passed

operating tests in which it was cooled by gaseous helium to temperatures of 20 to 30 kelvin. Although many steps remain to be completed before the SHiP magnet is ready, these are promising tests that open up prospects for this technology both at CERN and in industry.

“Such a magnet could consume up to 100 times less electrical power than an ordinary superferic magnet,” says Arnaud Devred, who is carrying out the project with a team from CERN’s Magnets group. “In the longer term, we could, for example, consider retrofitting certain magnets with MgB2 coils in order to reduce their electricity consumption. This project therefore represents a

great way of showcasing the technological developments for the HL-LHC.”

The superconducting links of the HL-LHC are attracting a lot of interest because they use high-temperature superconductors, whose large-scale use would allow significant energy savings in many areas, including in our everyday lives. Thanks to this highly innovative development, the scope of this technology can be extended to include electromagnets. The SHiP spectrometer magnet could be one of the first applications.

Corinne Pralavorio

Accelerator Report: No summer holidays for the accelerator complex

CERN’s accelerators show very good availability rates: beams are in place

Since [the technical stop in June](#), Linac4 has been running quite smoothly, delivering beam to the PS Booster with good availability, of 98.7%. Despite a small water leak from the cooling system in one of the PS Booster quadrupole magnets, beam availability remains high, at 94%. The leak, though small, is being carefully managed by diverting the water outside the magnet to prevent further issues. Continuous monitoring is in place, using a camera and data from the water-cooling station. As long as the leak does not worsen, operations will proceed as usual until the end of the 2024 run. If the leak increases significantly, a spare magnet is ready for installation, which would require a beam stop of several days.

As mentioned in [a previous Accelerator Report](#), physics at ISOLDE started on 8 April. Since then, approximately 20 different experimental runs have been conducted at the low-energy experimental stations, using various isotopes produced by impinging the high-intensity PS Booster proton beam on different types of targets. In parallel, the [HIE-ISOLDE](#) superconducting linear accelerator started the cooldown and conditioning of its 20 accelerating cavities mounted in four cryomodules and, last week, the first beams were accelerated to set up the experiments downstream. On 12 July, the physics campaign using post-accelerated isotope beams will start

and continue until the end of the 2024 ISOLDE run, scheduled for 25 November.



Five superconducting cavities mounted in a cryomodule. The picture was taken in a clean room before the insertion of the cryomodule into its cryostat. (Image: CERN)



The HIE-ISOLDE linear accelerator with its four cryomodules. (Image: CERN)

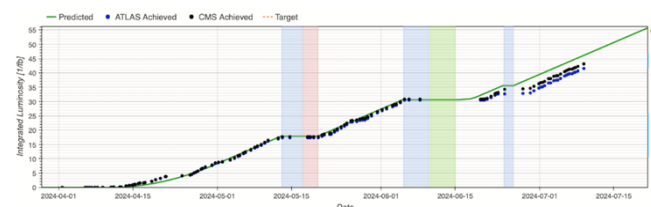
On the SPS side, following the [successful exchange of a magnet](#) on 18 June, the SPS resumed beam delivery for its usual clients: the North Area and the LHC. However, on 25 June, the SPS operators received an alarm indicating that some magnets in the SPS were overheating. The magnets are equipped with a magnet protection system that prevents them from being powered when their temperature rises too high, which also stops beam production.

The experts discovered that a blockage in the water-cooling circuit was causing the overheating. The circuit was unblocked and refilled on 26 June, allowing beam production to resume. The blockage was caused by pieces of rubber that had remained in the circuit following [an incident earlier this year](#).

Since resolving this latest and – hopefully – last issue, the SPS has had very good beam delivery, achieving beam availability of 97% for the LHC and 93% for the North Area, making up for some of the previously lost beam time. Additionally, the HiRadMat run, initially scheduled from 1 to 5 July, was already successfully completed by 2 July, allowing additional beam time for the North Area experiments.

The LHC resumed beam production on 28 June, after bringing forward some machine

development (MD) activities to give the ATLAS experiment a chance to recover from a cryogenic cooling issue. Since then, the luminosity production has been good, although the beams have often been dumped prematurely due to various unrelated and mostly minor technical issues. Despite this, the LHC had a machine availability of 70% last week, with the time spent in colliding-beams mode (i.e. stable beam ratio) of 51.3%, slightly above our target of 50%.



The integrated luminosity forecast (green line) with the achieved integrated luminosity for ATLAS (blue dots) and CMS (black dots). The difference between the two experiments is mainly due to ATLAS's cryogenic cooling issue. The fact that both are below the green line is mainly due to the advanced MD activities and the technical issues encountered, resulting in shorter stable beam periods and more frequent filling. Physics time will be recovered later, as the next MD block will be shorter. (Image: CERN)

Rende Steerenberg

CERN70: The end of the alphabet

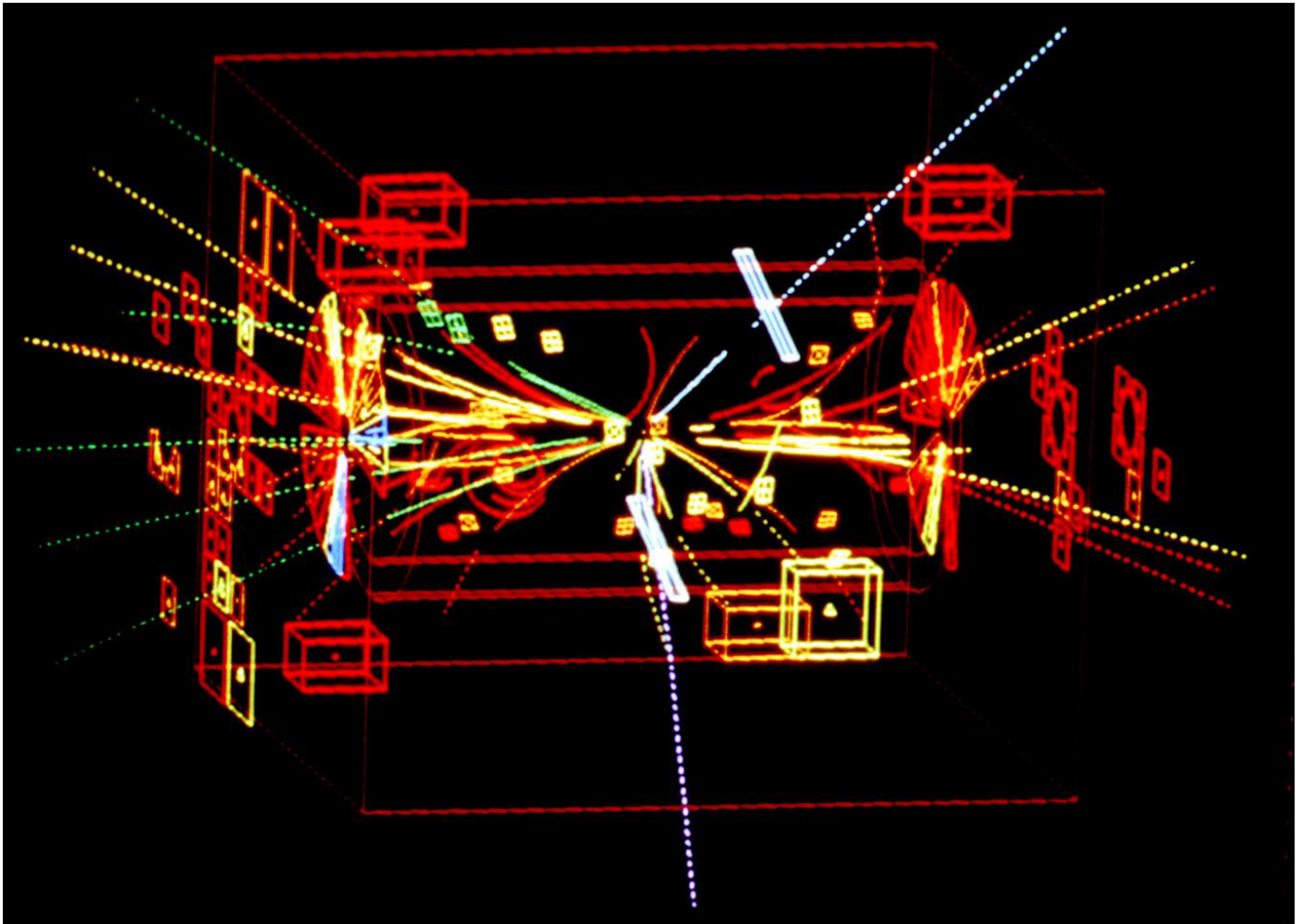
Part 13 of the [CERN70](#) feature series. Carlo Rubbia's name is closely related to the discovery of the W and Z particles at CERN

In 1983, CERN reached the end of the alphabet when the Laboratory announced the discovery of the long-sought [W](#) and [Z](#) particles. The announcement was so momentous that, the following year, the two scientists behind the discovery received the [Nobel Prize in Physics](#). In 1984, [Carlo Rubbia](#), the instigator of the conversion of the [Super Proton Synchrotron \(SPS\)](#) into a proton–antiproton collider and spokesman of the [UA1 experiment](#), and Simon van der Meer, who invented the stochastic cooling technology vital to the collider's operation, received the award from the Nobel Foundation.

To understand the importance of the discovery, one has to go back to three previous Nobel laureates. In the 1960s, [Steven Weinberg](#), [Abdus Salam](#), and [Sheldon Glashow](#) had proposed that the electromagnetic force and the weak force – which is responsible, for example, for some types of radioactivity – were both manifestations of a single interaction. According to this theory, three heavy particles, the W^+ , W^- and Z^0 bosons, are responsible for the “electroweak” force between particles.

First evidence of this unified force had already been found at CERN in 1973 by the [Gargamelle](#) experiment. But there was still no trace of the famous W and Z bosons, which were too heavy to be produced by the accelerators that existed at that time. The boson hunters were certainly not short of ideas, however. In 1976, David Cline, Peter McIntyre and Carlo Rubbia put forward the idea of converting CERN's largest accelerator,

the SPS, which had just come into operation, into a proton–antiproton collider, which would create sufficiently high energy levels to produce the long-awaited bosons.



The first Z event recorded by the UA1 experiment on 30 April 1983. (Image: CERN)

It took only three years to transform the accelerator, between 1978 and 1981, during which time two experiments UA1 and [UA2](#) were developed. The two detectors had the same main objective, the hunt for the W and Z bosons, but they differed in many ways. UA1, led by Carlo Rubbia, was the first of the large-scale collaborations, consisting of around 130 physicists. Its detector, which was huge for the time, weighed no less than 2000 tonnes. It had also been set numerous scientific objectives. UA2, which was designed specifically to search for the W and Z particles, was ten times smaller, and brought together a group of some 50 scientists.

The SPS recorded its first proton–antiproton collisions in July 1981. At the end of 1982, around one million events potentially providing evidence for the W and Z bosons had been recorded. On 21 January 1983, the UA1 team announced the discovery of the two W particles, which was confirmed by UA2. Ten times more W bosons were produced during the spring. In May, CERN announced the discovery of the third intermediate boson, the Z^0 .

Beyond its significance as a physics result, the discovery of the W and Z bosons was fundamental in reinforcing the conviction within the scientific community of the need to build the [Large Electron Positron \(LEP\)](#) accelerator, which came into service in 1989. LEP was a veritable Z factory – producing some 20 million Z particles – before becoming a W factory, and so measured the properties of both bosons with extreme precision.

Recollections

Carlo Rubbia's name is closely related to the discovery of the W and Z particles at CERN. In 1984, he was awarded the Nobel Prize in Physics, together with Simon van der Meer, for the work he had done as head of the UA1 collaboration. During his mandate as Director-General of CERN from 1989 to 1993, the Large Electron Positron collider was inaugurated and the four LEP experiments gave their first results.

"I came to CERN for the first time in 1960. Even now, I feel as motivated and enthusiastic for institutionalised international cooperation as I was on the first day, at the time when such an innovative concept, so popular nowadays, was essentially unknown.

CERN was born by a remarkable group of founders, at the threshold of a renewed European prosperity, to promote pure science on a global European scale and to create trust and unity between people of different countries, traditions and mentalities, after the disasters of the Second World War.



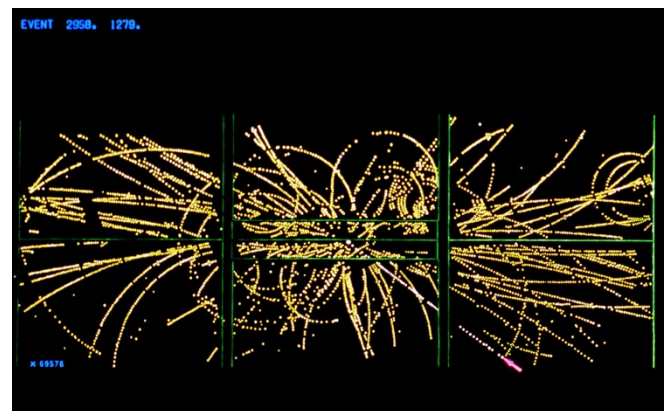
Carlo Rubbia (left) and Simon van der Meer celebrate being awarded the Nobel Prize in 1984. (Image: [CERN](#))

The originality of the CERN accelerator community has been evident from the beginning, with the realisation of the first strong-focusing proton accelerator by John Adams, then in his thirties, and by the remarkable team he had built up. He had the courage of cancelling the already approved 10 GeV weak focusing for a totally innovative 25 GeV

Proton Synchrotron. The next adventure was the realisation of the Intersecting Storage Rings, the ISR, which acted as a "reference" for all the subsequent projects, and which was made possible by the vision of the then Director-General, Viki Weisskopf.

However, the physics results, mostly because of non-optimal choices in the instrumentation at those times, were probably, in retrospect, below what they should have been. But the many accelerator innovations, learned with the ISR, allowed features for the SPS that were far more advanced than the ones at Fermilab in the United States, a relatively conventional accelerator, thus making possible a subsequent fast conversion into the SPS collider. An excellent accelerator, the SPS already had all the basic features to become a collider. All that was needed was an "industrial" antiproton source, at a time when even a handful of antiprotons was considered a success. This required overcoming the so-called Liouville theorem, amplifying early studies performed at the ISR by Wolfgang Schnell and Simon van der Meer. Enthusiasm and drive were so high that this jewel, the Antiproton Accumulator was built in less than two years!

The marvellous chain of new accelerators, SPS first and LEP later, have been complemented by the development at CERN of a number of equally revolutionary instrumentation technologies, such as the wire and drift chambers, the calorimeter (Herwig Schopper), and so on, without which all important discoveries in the field world-wide would have been impossible and for which Georges Charpak well deserved his Nobel Prize.



The discovery of the W particle in the UA1 detector from the October–December 1982 run of the proton–antiproton collider. (Image: [CERN](#))

The discovery of the W and the Z, promptly recognised by the Nobel Committee, was then so to speak “the tip of the iceberg” of a marvellous and unique adventure in which so many remarkable people from so many different countries, several of whom unfortunately no longer with us, have proudly contributed.

In order to assess the importance of what has been accomplished by the CERN teams in those days, it is sufficient to point out that until the LHC, the world’s hadronic high-energy frontier was dominated by two proton–antiproton colliders, the first entirely developed at CERN and the

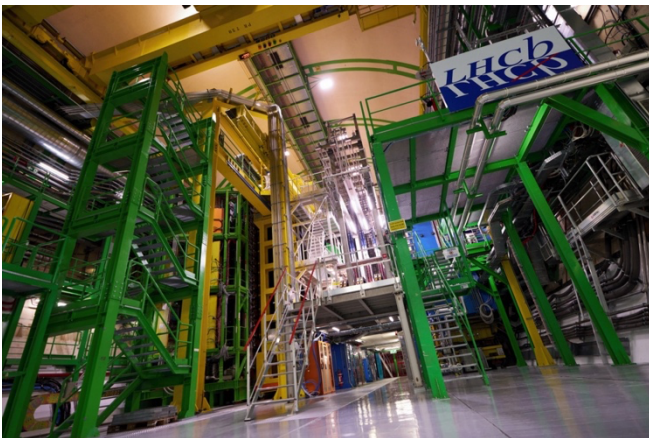
second incremented in energy with the Tevatron at Fermilab.

We have tremendously enjoyed this epoch of CERN: we are proud of what we have been able to accomplish. To quote Simon van der Meer: “If they have some idea – however crazy it is – they should check up on it. Once in a hundred times it will turn out to be a good idea.””

This interview is adapted from the 2004 book “Infinitely CERN”, published to celebrate CERN’s 50th anniversary.

News from LHCP: LHCb investigates the rare $\Sigma^+ \rightarrow p\mu^+\mu^-$ decay

The rarest hyperon decay ever observed



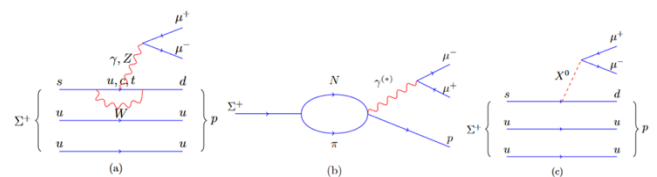
The LHCb detector. (Image: M. Brice/CERN)

The LHCb collaboration reported the observation of the hyperon $\Sigma^+ \rightarrow p\mu^+\mu^-$ rare decay at the [XV International Conference on Beauty, Charm, Hyperons in Hadronic Interactions \(BEACH 2024\)](#) in Charleston, South Carolina, USA. A hyperon is a particle containing three quarks, like the proton and neutron, including one or more strange quarks.

Rare decays of known particles are a promising tool for searching for physics beyond the Standard Model (SM) of particle physics. In the SM, the $\Sigma^+ \rightarrow p\mu^+\mu^-$ process is only possible through “loop diagrams”: rather than the decay happening directly, intermediate states need to be exchanged in a “loop”, as illustrated in diagrams (a) and (b) below.

In quantum field theory, the probability of such a process occurring is the sum of the probabilities of all possible particles exchanged in this loop, both

known and unknown. This is what makes such a process sensitive to new phenomena. If a discrepancy between the experimental measurement and theoretical calculations was observed, it could be caused by a contribution from some unknown particles. These particles could either be exchanged in the loop or mediate this decay directly, interacting with the quarks and then decaying into a pair of muons. In the latter case, shown in diagram (c) below, the new particle would leave a footprint in the properties of the two muons.



Feynman diagrams illustrating the $\Sigma^+ \rightarrow p\mu^+\mu^-$ decay in the Standard Model (diagrams a and b) and with a new X^0 intermediate particle (diagram c). (Image: LHCb)

Studies of the $\Sigma^+ \rightarrow p\mu^+\mu^-$ decay were especially exciting thanks to a hint of a structure that had been observed in the properties of the muon pair in 2005 by the HyperCP (E871) collaboration. With only three events, the structure was far from conclusive, and the LHCb study was expected to shine some light on the situation.

Ultimately, LHCb data does not show any significant peaking structures in the dimuon mass region highlighted by HyperCP, hence

disconfirming the hint. The new analysis does, however, observe the decay with high significance, and precise measurement of the decay probability along with other parameters will

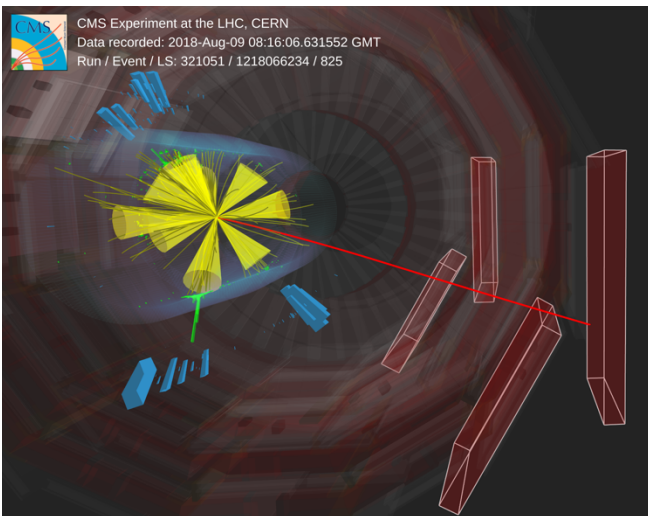
follow, allowing further searches for discrepancies with SM predictions.

Read more in the LHCb [presentations](#) at BEACH as well as in the conference [note](#).

Piotr Traczyk

News from LHCP: Going the extra mile to squeeze supersymmetry out of CMS data

Re-analysing LHC Run 2 data with cutting-edge analysis techniques allowed CMS physicists to address an old discrepancy



A candidate collision event with two top quarks and multiple reconstructed jets (shown as yellow cones). (Image: CMS collaboration)

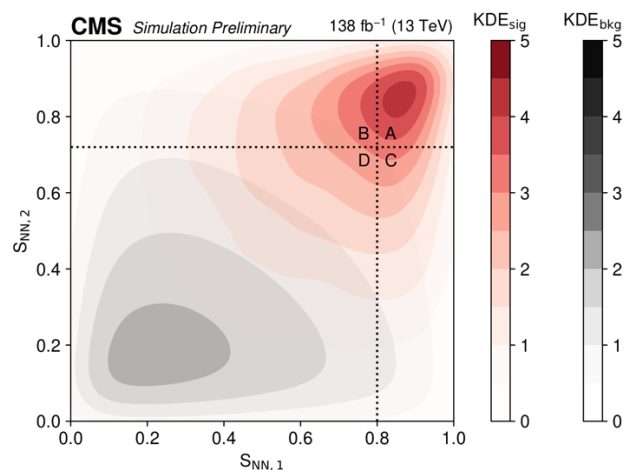
Supersymmetry (SUSY) is an exciting and beautiful theory that answers some of the open questions in particle physics. It predicts that all known particles have a “superpartner” with somewhat different properties. For example, the heaviest quark of the Standard Model, the top quark, would have a superpartner called the top squark, or simply the “stop”. In 2021 the CMS collaboration analysed the entire set of collision data collected from 2016 to 2018 and found features suggesting that it might contain stop particles. In that case, “might” meant that there was less than 5% chance that data containing only known particles could look like what was observed. Instead of waiting many years to collect more data with the hope of reproducing this behaviour, the CMS collaboration decided to reanalyse the same data with upgraded analysis techniques.

The new analysis looks for the simultaneous production of pairs of stops. Each stop decays into

a top quark accompanied by several lighter quarks or gluons, which then form bound states known as hadrons, ultimately creating clusters of particles reconstructed in the detector as “jets”. The signal footprint is therefore two top quarks and multiple jets. What makes the analysis challenging is that a very similar footprint is produced by one of the most common Standard Model processes in the LHC: the pair production of top quarks. Top quark production with many accompanying jets is a process that is difficult to accurately simulate, so to have a reliable determination of this background, it must be estimated from observed data.

A commonly used method of estimating backgrounds from data is called the “ABCD method”. It requires two uncorrelated observables that can discriminate between signal and background. The data set can then be divided into four regions (A, B, C and D) depending on the value of each observable being “signal-like” or “background-like”. The subdivision then provides a region dominated by the signal, a region dominated by backgrounds and two intermediate regions. The key feature of the ABCD method is that, following the mathematics of probabilities for independent events, one can estimate the background in the signal-dominated region using only the information from the other regions. The problem with using this method for the stop search is that all simple variables are correlated in this search, making the method invalid. To overcome this issue, CMS physicists have implemented an innovative approach based on advanced machine-learning techniques to determine two variables with a minimal level of

correlation. These two variables are then used to divide the data into the four aforementioned regions. The figure below shows the correlation between the two variables for the signal and the background and demonstrates that the signal mostly lies in region “A”.



Distributions of signal (red) and background (grey) in the four (A, B, C and D) regions, defined based on two uncorrelated variables (SNN1 and SNN2) determined using machine learning. (Image: CMS collaboration)

Using this novel method, the CMS collaboration was able to accurately predict the dominant background in this analysis from observed data, without relying on simulations with large uncertainties associated with the modelling of the jet multiplicity distribution. This resulted in a large gain in analysis sensitivity. If the signal hinted at by the 2021 analysis was real, it would now have been observed without any doubt. The fact that a signal was not seen in this analysis implies that, in specific SUSY scenarios, a stop decaying ultimately to top quarks and jets must have a mass greater than 700 GeV. With a much more sensitive analysis method in place, the physicists are now eagerly looking forward to analysing the data of the ongoing LHC Run 3 to go even further and to find where Nature hides its answers.

Read more in the CMS [Physics Analysis Summary](#).

CMS collaboration

How can I use CERN’s Open Science Office?

Publishing in open access, managing research data and open source software, and more: CERN’s Open Science Coordinator, Anne Gentil-Beccot, explains how the CERN community can benefit from the Open Science Office

Why should I care about open science?

The aim of open science is to make scientific research more accessible, transparent and efficient for the benefit of scientists and society. It includes making the products of research openly available – i.e. providing open access to research publications and sharing research data and open-source software and hardware – but also covers effecting cultural change in scientific processes to ensure that the production of knowledge is inclusive, sustainable and equitable.

What does CERN’s Open Science Office do?

The Open Science Office answers questions, provides guidance and connects the CERN community with experts and resources. It organises CERN [open science governance meetings](#) and plans to organise training courses and workshops in the future. It was created in

2023, following the release of CERN’s 2022 [open science policy](#), and will publish CERN’s first open science report in 2025.

Why is open science important to CERN?

We are lucky at CERN that our founding Convention gave us an early mandate for open science, and it has long been the norm within particle physics to share research results openly. With the digital transformation enabled by the World Wide Web (released by CERN in open source), CERN has pioneered a range of open science activities and services, making our Laboratory a world leader in open science. In recent years, government agencies and research funders from Member States and beyond have recognised the value of open science. This recognition is reflected in increased requirements

to demonstrate open science practices when submitting funding applications.

How can I contribute to open science?

You are already doing so when you submit your preprints to CDS or [arxiv](#), publish your research papers in open access in line with our open access [policy](#), make your experimental data accessible at [HEPdata](#) or through the CERN [Open Data Portal](#) or produce open source software. By sharing your work under licenses that let others use it, study its source code, redistribute it and share improvements, you are helping to promote transparent and inclusive research practices at CERN and beyond.

How can the Open Science Office help me?

Contact the [Open Science Office](#) with any questions you might have. We offer guidance on [open access publishing options](#) for your papers, as well as advice on [creating Data Management](#)

[Plans](#) (DMPs) to comply with funders' requirements, and much more.

I want my software or hardware to be open source. How can the Open Science Office help?

The newly launched [Open Source Programme Office](#) serves the CERN community with answers and insights regarding open source issues.

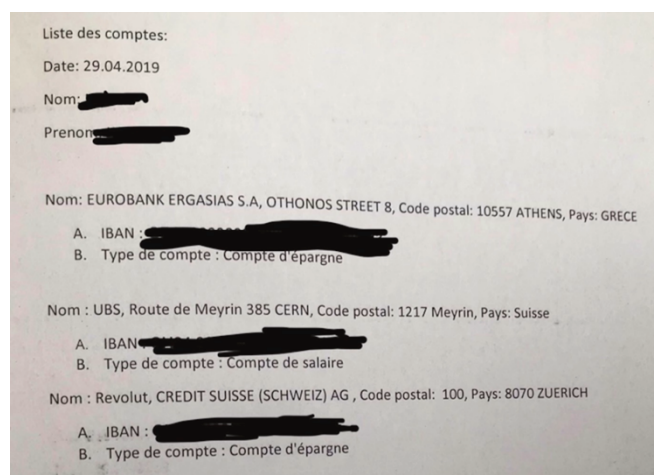
How can I find out more about open science at CERN?

You can consult the [CERN open science website](#), which includes details of upcoming events and training courses. You can also register for the [Open Science Practitioners Forum](#) e-group and join its regular meetings to have a say in CERN's open science strategy.

For more questions, contact the CERN open science team at os-office@cern.ch

Computer Security: Don't print naked

If you're interested in what's going on at CERN, in professional projects and plans, or in private problems and parties, hanging out at one of CERN's printers is a very effective approach (but, please... don't!). Too many people are still printing confidential documents without caring that they might be read by third parties hanging around – see for example the image below of a document found on one of CERN's printers a few years ago...

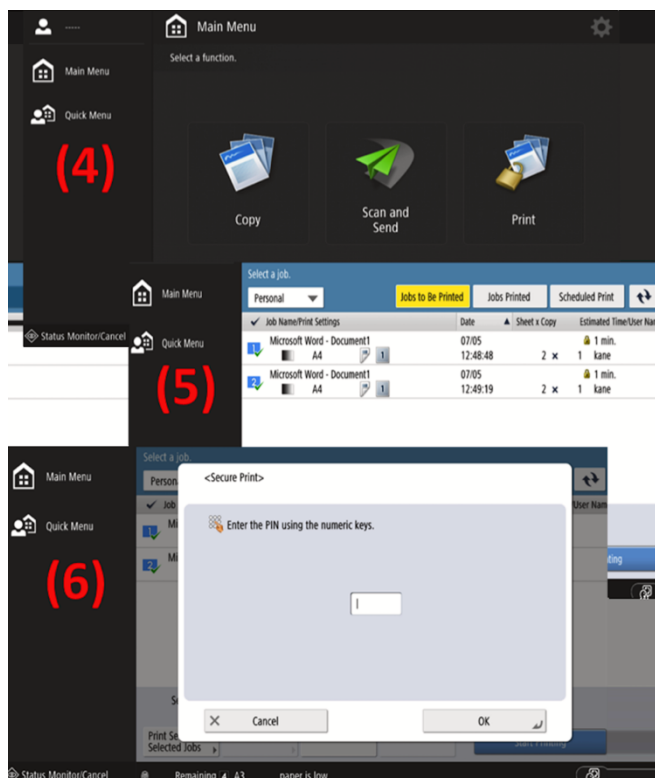
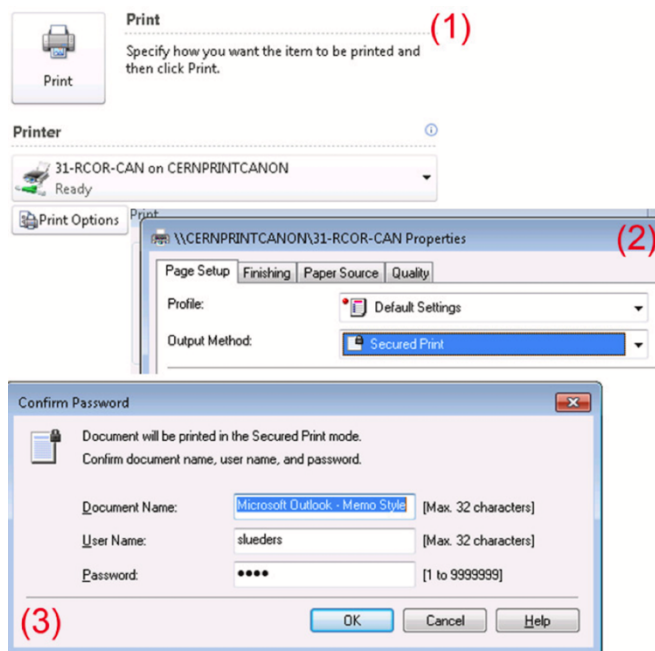


This shouldn't happen. Confidential documents, documents with sensitive content, personal information and private emails should be properly protected – even when they're converted to paper format.

The CERN printing infrastructure is capable of ensuring the confidentiality of your documents: you can send a print job in such a way that it won't be printed until you input a PIN code at the machine. So the next time you need to print such a document, go to "Printer Properties" (1) and select the "Secure Print" output method (2). Once you hit the "Print" button, you'll get another dialog box asking you to provide a PIN number (3). Below, you can see screenshots of the different steps on the Windows platform. Instructions for other operating systems can be found in this [ServiceNow knowledge base article](#).

After you've provided a PIN, your document will be queued on the printer of your choice. For easy PIN input on the printer, we recommend you use any Canon machine. The printer will hold your document for the next 12 hours (4 hours on some

models). Once you're at the printer, hit the "SCAN/PRINT" button, choose "Secure Print" (4), select the jobs you want to print (5), hit "Print", and punch in your PIN (6). (Note: this sequence may vary between models.)



Your document is only printed after you provide the correct PIN. And as you're now standing beside the printer, you can be confident that your confidential document is being handled confidentially. And, even better, you can set this as your default printing preference and ensure none of your printouts go walkabout again!

See [these knowledge base articles for more details](#) – including short video tutorials. In addition, you can [download this poster for the corresponding awareness campaign](#) by the Office of Data Privacy (ODP) and pin it up (close to) your printer(s).

So, don't let confidential information leak... use the "secure print" function on your nearest preferred printer. And: respect the environment. Don't print unless necessary. Confidential documents that are never printed are less likely to slip out of your control!

Computer Security team

More information on data privacy at CERN can be found on the [ODP website](#) or obtained from your [Departmental Data Privacy Coordinator](#).

Official News

Members of the personnel shall be deemed to have taken note of official news published for the CERN community. Reproduction of all or part of this information by persons or institutions external to the Organization requires the prior approval of the CERN Management. The full list of official news is available here: <https://cern.ch/official-news>.

Updated factors for anticipated retirement pension

During the public information [sessions](#) in January 2024, the Pension Fund informed you of changes to the Pension Fund Rules and Regulations with respect to updated factors due to be implemented during 2024.

We would like to remind you that the updated factors for anticipated (early) retirement will be

effective from 1 July 2024 and are available on our [website](#) and in the [Pension Fund Rules and Regulations](#) (Art. II 2.05). Factors for existing benefits under anticipated retirement are not affected by this change.

Pension Fund
(pension-benefits@cern.ch or +41 22 767 88 11)

Procedure for obtaining visas for Switzerland and France – Signature rights

In accordance with the Status Agreements with CERN, Switzerland and France facilitate the entry of members of the Organization's personnel on to their territories. Where relevant, detailed procedures for obtaining visas apply.

Within the framework of those procedures, **only** the following individuals are authorised to initiate the *Note verbale* procedure as well as to sign the Official Invitation Letters and the *Conventions d'accueil*.

1. Kirsti ASPOLA (EP – CMO)
2. Maria BARROSO LOPEZ (IT – RM)
3. Catherine BRANDT (DG – DI)
4. Michelle CONNOR (TH – GS)
5. Rachelle DECREUSE-MICHAUD (EN – PAS)
6. Cindy DENIS (EP – AGS)
7. Gaëlle DUPERRIER (EP – AGS)
8. Nathalie GOURIOU (EP – AGS)
9. Donia GRANDCLAUDE (HR – CBS)
10. Nathalie GRÜB (EP – AGS)
11. Cassandra HEIGHTON (BE – HDO)
12. Georgina HOBGEN (SY – AR)
13. Jeannette KOTZIAN (ATS – DO)
14. Lucie MAINOLI (SY – AR)
15. Tania PARDO (EP – AGS)

16. Maria QUINTAS (HR – CBS)
17. Kate RICHARDSON (EP – AGS)
18. Christophe SCHAEFER (IR – DS)
19. Emmanuel TSESMELIS (IR – DS)

The French and Swiss Authorities will reject any request signed by a person who is not on this list.

We would like to remind you that in accordance with the memorandum of 7 December 2000 issued by the Director of the Administration, (ref. DG/DA/00-119), "*the Organization shall not request any legitimisation document (or residence permit) or visa from the Host States for persons registered as EXTERNAL*" (people who do not hold a contract of employment, association or apprenticeship with CERN).

We would also like to remind you that those coming to CERN should find out **in good time and prior to their arrival** about the conditions of entry to Switzerland and France applying to them and ensure that they obtain the requisite visa, where applicable, **in the country in which they are habitually resident**. The application must be completed 6 months to 3 weeks before the travel.

Useful information can be obtained from the Swiss and French diplomatic representations abroad, as well as from the following Web pages:

- CERN Admin e-guide: <https://admin-eguide.web.cern.ch/en/procedure/visas-entry-and-stays-host-states>;
- Swiss State Secretariat for Migration: https://www.sem.admin.ch/sem/en/home/themen/einreise/merkblatt_einreise.html;
- Swiss permanent Mission in Geneva: <https://www.eda.admin.ch/missions/mission-onu-geneve/en/home/manual-regime-privileges-and->

immunities/introduction/manual-visas/schengen-visas-entry-exit-travel.html;

- French Ministry for Europe and Foreign Affairs and Ministry of the Interior: https://france-visas.gouv.fr/en_US/web/france-visas.

The Authorities of the Host States have informed the Organization on a number of occasions that they insist upon **scrupulous compliance** with visa legislation.

Host State Relations Service
<http://www.cern.ch/relations/relations.secretariat@cern.ch>
Tel.: 75152

Announcements

Upcoming events

Check out this curated list of upcoming events for the CERN community

14 Jun–14 Jul | At CERN | R1 | Novae | FR
[Euro 2024 matches on the big screen](#)

2 Jul–2 Aug | Knowledge Sharing | Main Auditorium | Summer Student Programme | EN
[Summer student lecture programme 2024](#)

2–31 Jul | Knowledge Sharing | IT Amphitheatre | CERN openlab | EN
[CERN openlab summer student lecture programme 2024](#)

2–12 Jul | Knowledge Sharing | Palais des Nations | UN Geneva, CERN & ITU | EN
[Exhibition: Not a Woman's Job? Changing Faces of Science and Tech](#)

8–19 Jul | Physics | CERN | Theory Institute | EN
[Lattice@CERN 2024](#)

3 Aug 14:00 | At CERN | R1 Terrace, Meyrin site | Music Club | EN & FR
[Music on the Lawn](#)

13 Aug 10:00 | Knowledge Sharing | CERN Library | Archives, Library and Open Science events | EN
[Meet the authors of "New Challenges and Opportunities in Physics Education"](#)

Registration now open

17 Sep 18:00 | At CERN | R1 lawn | CERN70 | EN & FR
[CERN70 Community Event](#) (more [CERN70](#) events)

22 Sep–5 Oct | Accelerators | Santa Susanna (ES) | CERN Accelerator School | EN
[Introduction to Accelerator Physics](#)

11 Oct 14:00 | Knowledge Sharing | Online | HR Learning and Development | EN
[L&D Micro-talk 12 – Demystifying creativity](#)

27 Nov–6 Dec | Experiments | CERN | DRD1 | EN
[DRD1 Gaseous Detectors School](#)

This is a curated list of events relevant to the CERN community. More events are available here: home.cern/events

Indico also shows ALL events happening [today](#), [this week](#) and in a [calendar view](#).

If you would like your event to appear on an upcoming Bulletin events list, please contact bulletin-editors@cern.ch

Invitation to the CERN70 community event



In celebration of our Laboratory's 70th anniversary, we are thrilled to invite you to a special event. This occasion honours the scientific excellence, spirit of international collaboration, openness and inclusiveness that have driven our community to the forefront of scientific discovery and innovation for seven decades.

The celebration will take place on the Meyrin site on **Tuesday 17 September 2024**, from **6 p.m. to 10 p.m. CEST**, on the large lawn in front of Restaurant 1 and its surroundings.

All holders of a [valid CERN access card](#), including the family members of the personnel (spouse, registered partner, children), are warmly invited to join the event. Please note that visitor passes are not permitted on that day.

Prepare for an evening of diverse entertainment featuring musical performances, special guests

and many surprises. A variety of world street food and drinks will be available to everyone, free of charge.

[Register on Indico](#) (please log in to see the registration link on the menu bar on the left) at your earliest convenience but **no later than 31 August**. By registering, you will receive a ticket and QR code to **collect the food and drink vouchers at the indicated [days, times and locations](#)**. People who do not have a CERN account will be contacted via email shortly with instructions on how to register.

More practical information about site access, parking and shuttle services will be available in early September and communicated to all registered participants.

We look forward to celebrating this important milestone with you!

Design the official T-shirt for the CERN70 Community Event!

Calling all CERN community members: the CERN70 team is launching a competition for the design of the official T-shirt for the CERN70 Community Event

As part of CERN's 70th anniversary celebrations, the CERN70 team is holding a competition for the design of the official T-shirt for the CERN Community Event, which will take place on 17 September (don't forget to register!).

All the designs entered in the competition will be displayed in the Main Building (Building 500) from 30 July to 13 August 2024.

The winning design will be chosen based on several criteria (in particular, visual quality, originality and representation of CERN values) by

a jury composed of seven people, including members of the CERN70 team and CERN communication professionals.

The official T-shirt will be sold at cost price during the event on 17 September.

How to enter:

- The competition is open to **all CERN badge holders**.
- Send your designs to CERN70.T-shirt@cern.ch **before 5.00 p.m. (CEST) on 29 July 2024**.

- Participants must **strictly** respect the [technical requirements](#) (including regarding use of the CERN logos).
- The winning design will be published on the [CERN70 website](#) and in the CERN Bulletin, and the winner will receive a 100 CHF voucher for the [CERN Shop](#) and two T-shirts.

All the details are available [here](#).

Good luck!

A streamlined procedure for obtaining dosimeters

The HSE-RP Dosimetry service launches a new optimised dosimeter request process in ServiceNow

The [Dosimetry service](#) supplies some 11 000 dosimeters annually for gamma and neutron radiation monitoring of individuals accessing CERN radiation areas during the course of their professional activity at CERN (radiation workers). The people monitored include employed and associated members of the personnel, contractors and temporary personnel. To obtain a dosimeter, radiation workers must complete the relevant radiation protection safety courses and provide the requested documents.

The process has now been streamlined in ServiceNow with a simple and straightforward form in order to ensure that all steps are duly completed prior to collecting the dosimeter at Building 55. This process will guide radiation workers on the different steps to complete prior to collecting their dosimeter, thus reducing waiting times and optimising their experience. It will also help the Dosimetry service better anticipate and manage the volume of requests.

If you need a dosimeter or wish to request a dosimeter for a third party, whatever the CERN status and the type of dosimeter needed, please ensure you first complete the [ServiceNow form online](#). Once submitted, we will process your request and inform you as soon as the dosimeter is ready for collection at Building 55*. The form may also be accessed via the [CERN Campus App](#), under “Campus Life”, “Dosimetry”.

If you already own a dosimeter of the DIS type, the process remains unchanged: you will continue to receive annual email reminders to exchange it.

The HSE-RP Dosimetry service

**Note: All services located at Building 55, including the Dosimetry service, are due to move to new premises in Building 33 on 1 August. Further information will be communicated in due course.*

Next blood donation campaign on 16 July

At the last blood donation campaign on 26 March, 148 donors came along to give blood, of which 43 were first-timers. 96 donations were collected.

The Geneva [blood barometer](#), which is regularly updated, shows that blood stocks continue to run extremely low. CERN will hold a third blood donation session on Tuesday 16 July, from 8.30 a.m. to 4.30 p.m. in collaboration with

the Hôpitaux universitaires de Genève ([HUG](#)), in [Building 29](#).

Before coming to the Building 29 donation point, please make sure that:

- you are in good health and have no symptoms such as fever, cough, a cold or breathing difficulties;
- you are eligible to give blood:

- Complete the brief online [pre-donation questionnaire](#) (in French)
- Please consult this [information sheet](#) and complete this [questionnaire](#) issued by the HUG (but note that only the pre-donation conversation with the nurse or doctor on the day can confirm your eligibility).
- If returning from travel abroad or arriving from another country, perform the [travelcheck](#) on the HUG website (in French)
- If you have recently been vaccinated against the flu or COVID, ensure 48 hours have passed since receiving the vaccination.

- Please bring proof of ID with you on the day of the donation.

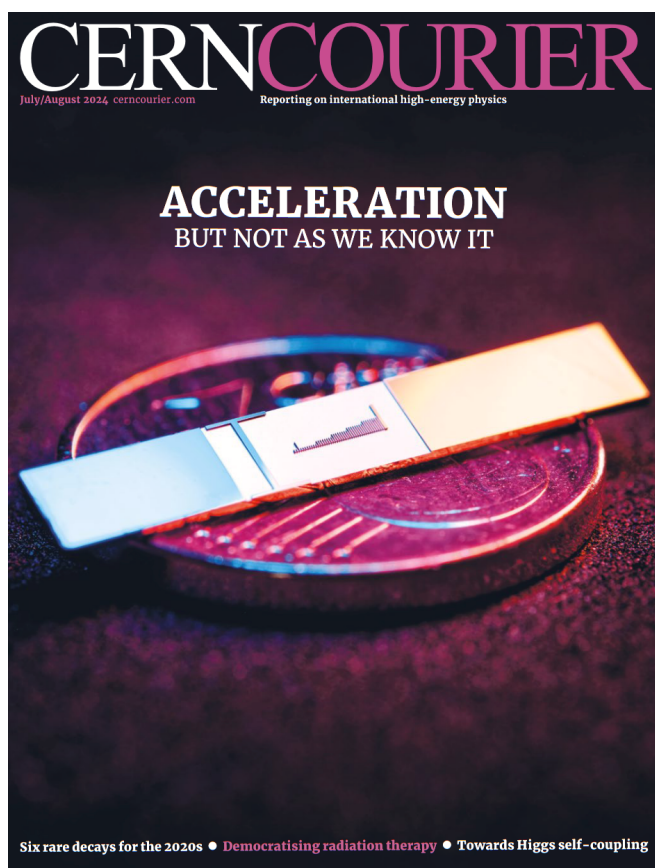
Save yourself time: pre-book your appointment with OneDoc

The possibility of pre-booking your appointment via the OneDoc platform is made available on this [link](#). Spontaneous donors will be able to come at their convenience without an appointment, mindful that there may be some waiting time on the day.

As a thank you, the HUG will be giving each donor a 10 CHF voucher, to be used in Novae's Restaurants 1 and 2 at CERN.

We hope to see many of you at the donation point!

The July/August issue of the CERN Courier is out



Insulators rather than conductors. Pillars rather than cavities. Microns rather than centimetres. Over the past 30 years, a handful of research groups have been quietly reimagining the electron linac in miniature. Their work is developing rapidly, with focusing, bunching and net acceleration over hundreds of optical cycles all now demonstrated “on chip”. This is acceleration, but not as we know it. Elsewhere in this issue: Andrzej Buras identifies the six rare decays that are best placed to probe beyond the energy frontier this decade; CALET studies cosmic-ray anomalies on the International Space Station; DUNE takes shape in Ray Davis Jr’s gold mine; how to increase access to medical linacs in low- and middle-income countries; ATLAS and CMS hunt the Higgs boson’s self-interaction; and how big is a neutrino?

[Read the digital edition of this new issue on the CERN Courier website.](#)

CERN Community: thanks for taking part in the CERN & Society Foundation anniversary game

See the winners and solution below

Thank you to all who participated in [the CERN & Society Foundation \(CSF\) anniversary game](#) for the CERN Community.

We hope you learned interesting things about the [CERN & Society Foundation](#) and its mission.

Congratulations to Karan Nautiyal (BE-ICS), Michael Hemmett (EP-UAT) and Thibaut Lang (EN-AA), who all had the correct answer and were selected in the prize draw. They each win a CERN & Society Foundation water bottle.

The answer of the game is 327 293. It is the sum of numbers that appear on-screen in the [CSF 10th anniversary impact video](#). (We also accepted 327 297, as some of you included the 4 from “Beamline 4 Schools” ;)).

Internal communication

Obituaries

Susan Lesley Bland (1943 – 2024)



Susan pictured with her beloved granddaughter Nora. (Image: Susan's family)

Susan Lesley Bland was born in 1943 in Leeds, UK. At the age of 16, she decided to stop her studies to support her younger brother, who was suffering from leukaemia and who died shortly after. She then joined her sister in Switzerland and entered CERN in 1962 at the age of 18.

Susan pursued a career in administration, where she gradually rose in the hierarchy to become responsible for the secretariats of the Proton Synchrotron (PS) division. At selection boards, her

insights into human nature were always much appreciated and her opinion was often decisive. She also made the first web version of the CERN administrative procedures and maintained its content for many years. Later, she was instrumental in archiving historically important PS documents.

Susan had two children, Sophie and Stéphanie, with her first husband, Serge Neboux, and their well-being was her main concern. She later met systems engineer Robert Cailliau, who would become her second husband. With him, she began a life punctuated by travels and projects, but above all she supported him in the journey of developing the web in the early 1990s.

In 2005, Susan retired after 43 years – constituting one of the longest ever careers at CERN. In her retirement, she became even more passionate about her hobby of genealogical research and participated actively in many community forums. Through this research, she was able to list no fewer than 60 000 people linked to her family.

Susan remained a pillar to us all until the end. We will miss her dearly.

Her family